Prabhavit Gupta



AI Engineer: Generative AI, RAG Architectures & Production LLM Systems

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Summary

AI Solutions & Automation Engineer with 3 years of hands-on production experience architecting, deploying, and maintaining production-ready Generative AI systems, RAG pipelines, and enterprise automation workflows. Proven track record turning complex enterprise requirements into scalable LLM applications, integrating multi-modal APIs, vector search datastores, and cloud infrastructure across European client environments. Deep practical experience implementing evaluation harnesses, managing token optimization, and establishing CI/CD and MLOps deployment standards. Holder of an EU Long-Term Residence Permit (permanent unrestricted EU work authorization across all EU member states, no sponsorship required); based in Venice, Italy and available immediately for remote freelance contracts and European on-site travel.

Technical Expertise

- **Generative AI & LLM Applications:** LLM APIs (OpenAI, Anthropic), RAG architectures, semantic retrieval, embeddings, vector indexing, prompt engineering, context management, structured JSON outputs, transcript evaluation harnesses.
- **Software Development & Integration:** Python, TypeScript, Node.js, RESTful APIs, webhook consumers, data ingestion and transformation pipelines, PostgreSQL schema modeling, Redis asynchronous queues.
- **Cloud & MLOps Practices:** Cloud environments (Azure, AWS), Docker containerization, CI/CD pipelines, Git version control, observability and trace monitoring, automated regression testing with Playwright.
- **Workflow Automation & Enterprise Delivery:** n8n orchestration, third-party system synchronization, SLA compliance, technical documentation, cross-functional stakeholder collaboration.
- **Languages & Work Authorization:** English (C1 Professional, fluent), Hindi (Native), Italian (Beginner). EU Long-Term Residence Permit (unrestricted EU work authorization across all EU member states, no sponsorship required).

Experience

FlowSpace.co **Mar 2024 - Aug 2026**
Lead Developer & AI Systems Architect *London, UK (remote)*

- Designed and operated production customer-facing AI agents for a marketplace platform serving 4,500+ active enterprise users, handling complex inquiries against live platform data and APIs.
- Implemented automated document and data parsing pipelines with LLM extraction, converting unstructured partner data into validated JSON schemas and expanding live platform inventory.
- Built human-in-the-loop escalation workflows and multi-step reasoning chains with strict tool constraints, ensuring zero ungrounded responses in customer-facing flows.
- Evaluated system prompts and tool schemas against real production transcripts, systematically lowering edge-case error rates and measuring response quality against ground-truth validation datasets.
- Maintained resilient backend integrations with TypeScript/Node.js and PostgreSQL, implementing asynchronous queues and automated retries for high-throughput reliability.
- Built automated end-to-end integration and regression test suites with Playwright, ensuring API contract validity and zero regressions across continuous release cycles.

Urlink **Apr 2025 - Jul 2025**
Project Manager & Solutions Developer (Freelance) *Barcelona, Spain (remote)*

- Led technical discovery and implementation of customized web solutions and automated data processing pipelines for client platforms.
- Built automated data ingestion and transformation services connecting third-party APIs to PostgreSQL datastores, maintaining high data freshness and integrity.
- Delivered interactive functional prototypes to validate user flows and technical requirements prior to sprint execution, accelerating cycle times.

Independent Technical Consultant **Aug 2023 - Feb 2024**
AI Automation & Integration Developer *Venice, Italy*

- Designed, deployed, and supported tailored LLM integrations, document-parsing pipelines, and webhook orchestration workflows for European clients using Python, TypeScript, and n8n.
- Replaced fragile manual data entry workflows with automated sync pipelines featuring structured error handling, retry policies, and alerting.
- Documented system architectures and delivered technical walkthroughs, ensuring smooth client adoption and high operational reliability.

Education

Ca' Foscari University of Venice **2020 - 2024**
BA in Philosophy, International Studies and Economics *Venice, Italy*

- Rigorous training in formal logic, analytical problem solving, systems analysis, and macroeconomic decision models.